



MIST- MEDICAL INSTITUTE FOR SCREENING TEST

MOCK - 3 PAPER - 1 DATED ON 26 MAY 2026

QUESTIONS

- Q1. A 28-year-old male is admitted following a high-velocity road traffic collision resulting in a direct rotational force applied to the right upper limb. Clinical examination reveals gross deformity, crepitus, and localized ecchymosis over the mid-diaphyseal region of the humerus. Radiographic imaging confirms a spiral fracture of the humeral shaft. Initially, distal motor function appears preserved; however, within hours, the patient develops a progressive motor deficit characterized by inability to extend the wrist, metacarpophalangeal joints, and thumb, accompanied by attenuated grip strength due to loss of wrist stabilization. Neurological assessment further demonstrates hypoesthesia localized to the dorsolateral aspect of the hand, particularly within the first dorsal intermetacarpal web space, while triceps function is relatively spared. The injured neural structure is known to course within the spiral groove of the humerus, intimately related to the periosteum and susceptible to entrapment or laceration in diaphyseal fractures. Which of the following clinical syndromes most accurately represents the expected neuromuscular deficit associated with this lesion?
- a) Flaccid paralysis of intrinsic hand musculature with pan-digital clawing
 - b) Upper brachial plexus palsy with loss of shoulder abduction and forearm supination
 - c) High radial neuropathy resulting in loss of wrist and digital extension with preserved partial elbow extension
 - d) Isolated ulnar neuropathy producing Froment's sign and hypothenar wasting
- Q2. A man with a malignant tumor in the upper rectum develops hepatic metastases. Which venous pathway most likely carried tumor cells to the liver first?
- a) Middle rectal vein → internal iliac vein → IVC
 - b) Inferior rectal vein → internal pudendal vein → IVC
 - c) Superior rectal vein → inferior mesenteric vein → portal vein
 - d) Internal vertebral venous plexus
- Q3. A patient suffers a lesion of the left lateral medulla. Which of the following findings is most likely?
- a) Ipsilateral loss of facial pain and contralateral loss of body pain
 - b) Contralateral loss of vibration and proprioception from the body
 - c) Ipsilateral spastic hemiparesis
 - d) Bilateral lower motor neuron facial paralysis
- Q4. A patient with long-standing hypertension develops concentric left ventricular hypertrophy. Which of the following changes in ventricular pressure-volume relationship is most likely?
- a) Increased end-diastolic compliance and rightward shift of the diastolic pressure-volume curve
 - b) Decreased end-systolic elastance due to systolic failure
 - c) Decreased ventricular compliance with upward shift of the diastolic pressure-volume curve
 - d) Increased stroke volume due to reduced afterload
- Q5. A 5-month-old infant is brought with recurrent episodes of severe fasting hypoglycemia, irritability, and poor feeding. On examination, there is marked hepatomegaly. Laboratory investigations reveal lactic



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acidosis, hyperuricemia, and hyperlipidemia. The child shows minimal rise in blood glucose after glucagon administration. Which of the following enzymes is most likely deficient?

- a) Glucose-6-phosphatase
- b) Debranching enzyme
- c) Glycogen phosphorylase
- d) Branching enzyme

Q6. A 10-day-old neonate is brought with poor feeding, lethargy, and recurrent vomiting. Parents report a characteristic odor in the baby's urine. On examination, the infant shows hypotonia and episodes of seizures. Laboratory studies reveal accumulation of leucine, isoleucine, valine. Which of the following enzyme deficiencies is most likely responsible for this condition?

- a) Branched-chain α -ketoacid dehydrogenase
- b) Pyruvate dehydrogenase
- c) Transaminase
- d) Hexokinase

Q7. A patient with chronic metabolic acidosis shows increased renal ammoniogenesis. Which of the following biochemical processes is most directly upregulated in proximal tubular cells?

- a) Conversion of glutamate to α -ketoglutarate with NADH consumption
- b) Deamination of glutamine producing NH_3 and bicarbonate
- c) Increased urea cycle flux in renal mitochondria
- d) Increased lactate production from pyruvate

Q8. A woman in labor receives a pudendal nerve block for pain control during the second stage of labor. The anesthetic is typically injected near which bony landmark?

- a) Ischial spine
- b) Pubic tubercle
- c) Greater trochanter
- d) Sacral promontory

Q9. A 50-year-old man presents with progressive weakness of the hand and difficulty in gripping objects. He also complains of numbness over the little finger and medial half of the ring finger. On examination, there is clawing of the medial two digits with hyperextension at the metacarpophalangeal joints and flexion at the interphalangeal joints. Interossei muscle wasting is noted, and finger abduction and adduction is impaired. The symptoms developed gradually after prolonged leaning on the elbow while working. Which of the following is the most likely site of nerve compression?

- a) Carpal tunnel
- b) Cubital tunnel
- c) Axilla
- d) Wrist extensor

Q10. A 38-year-old male presents with sudden onset deviation of the mouth to one side and inability to close his right eye. He also reports drooling of saliva and difficulty in blowing air through his cheeks. On examination, there is flattening of the nasolabial fold on the affected side. Taste sensation over the anterior



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tongue is reduced. There is no limb weakness. Which of the following nerves is most likely affected in this patient?

- a) Trigeminal nerve
- b) Facial nerve
- c) Glossopharyngeal nerve
- d) Vagus nerve

Q11. A 30-year-old male presents with a history of untreated facial infection involving the upper lip and nose region. He now complains of diplopia and inability to move his right eye laterally. On examination, the right eye is deviated medially, and abduction is restricted. There is associated periorbital edema and signs of cavernous sinus involvement. Considering the anatomical relations within the cavernous sinus, which of the following cranial nerves is most likely the first to be affected in this condition?

- a) CN III
- b) CN IV
- c) CN VI
- d) CN V

Q12. A 45-year-old chronic smoker found to have an apical lung mass compressing the sympathetic chain. Which of the following clinical findings is most characteristic of this condition?

- a) Ptosis, miosis, anhidrosis
- b) Mydriasis, exophthalmos, hyperhidrosis
- c) Ptosis, mydriasis, lacrimation
- d) Proptosis, diplopia, photophobia

Q13. A 26-year-old male presents with an intermittently reducible inguinoscrotal swelling that exhibits a positive expansile cough impulse. The swelling increases on Valsalva maneuver and reduces spontaneously in the supine position. On deep palpation, the hernial sac is found to enter the inguinal canal laterally to the vessels, traversing the entire canal and emerging through the superficial ring. There is no prior surgical history, and the defect is suspected to be related to persistence of an embryonic peritoneal diverticulum.

- a) Direct inguinal hernia
- b) Indirect inguinal hernia
- c) Femoral hernia
- d) Umbilical hernia

Q14. 66-year-old man undergoes radical prostatectomy for adenocarcinoma. Postoperatively, he has erectile dysfunction despite preserved penile blood flow and intact testosterone levels. Injury to which structure is the most likely cause?

- a) Pudendal nerve
- b) Pelvic splanchnic nerves
- c) Ilioinguinal nerve
- d) Genitofemoral nerve

Q15. A 22-year-old male is brought to the emergency department after a road traffic accident. He had a brief loss of consciousness followed by a lucid interval. A few hours later, he develops severe headache, vomiting, and progressive drowsiness. On examination, there is a temporal scalp swelling. CT scan reveals



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a hyperdense lesion in the temporal region. The fracture line is suspected over the pterion. Injury to which of the following vessels is most likely responsible for this condition?

- a) Internal carotid artery
- b) Middle meningeal artery
- c) Basilar artery
- d) Vertebral artery

Q16. A 19-year-old male is brought to the emergency department after a fall from a bike.. After a few hours, he develops severe headache, vomiting, and progressively decreasing level of consciousness. On examination, there are signs of raised intracranial pressure. CT scan shows a biconvex lens-shaped hyperdense lesion in the temporal region. What is the most likely underlying cause of this clinical presentation?

- a) Venous bleed
- b) Arterial bleed
- c) CSF leak
- d) Edema

Q17. A 35-year-old male presents with difficulty in walking following trauma to the lateral aspect of the knee. He is unable to dorsiflex the foot and has a characteristic high-stepping gait. Which of the following is the most common site of injury?

- a) Femur
- b) Fibular neck
- c) Tibia
- d) Ankle

Q18. A 42-year-old female presents with numbness, tingling, and burning sensation over the thumb, index, middle, and lateral half of the ring finger, which worsens at night. She also reports difficulty in gripping objects and weakness of thumb opposition. On examination, there is thenar muscle wasting and a positive Tinel's sign at the wrist. The condition is due to compression of a nerve within a fibro-osseous tunnel at the wrist. Under which of the following structures does this compression occur?

- a) Flexor retinaculum
- b) Extensor retinaculum
- c) Deltoid ligament
- d) Annular ligament

Q19. A newborn is brought to the pediatric clinic with weakness of the right upper limb following a difficult vaginal delivery with excessive traction on the neck. On examination, the arm is adducted, medially rotated, the elbow is extended, forearm pronated, and wrist flexed. The infant is unable to abduct or externally rotate the shoulder. These findings suggest injury to the trunk of the brachial plexus. Which of the following deformities is most characteristic of this condition?

- a) Claw hand
- b) Wrist drop
- c) Waiter's tip deformity
- d) Foot drop



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Q20. An elderly patient presents with progressive headache, confusion, and drowsiness several weeks after a minor head injury. There is no history of immediate loss of consciousness. On examination, he has mild hemiparesis. CT scan reveals a crescent-shaped hypodense collection over the cerebral hemisphere. Rupture of which of the following structures is responsible for this condition?

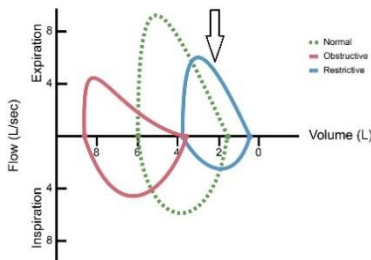
- a) Arteries
- b) Bridging veins
- c) Capillaries
- d) Sinuses

Q21. A newborn is found to have herniation of abdominal viscera into the base of the umbilical cord, covered by a translucent sac. Which embryologic defect best explains this anomaly?

- a) Failure of lateral body folds to close with bowel herniation through the abdominal wall, usually to the right of the umbilicus
- b) Persistence of physiologic midgut herniation into the umbilical cord
- c) Failure of cloacal membrane rupture
- d) Failure of neural tube closure in the lumbosacral region

Q22. A 52-year-old man presents with progressive breathlessness on exertion over the past 1 year. He has no history of smoking. On examination, heart sounds are normal and auscultation of the lungs does not reveal wheeze or crackles. Chest X-ray is unremarkable. Pulmonary function testing is performed, and the flow-volume loop (marked by arrow) shows a reduced lung volume with a preserved shape and no scooping of the expiratory limb. What is the most likely diagnosis based on this flow-volume curve?

Flow Volume Loops



© Lineage

Moises Dominguez

- a) Extra-parenchymal restrictive lung disease
- b) Parenchymal restrictive lung disease
- c) Obstructive lung disease
- d) Normal lung field

Q23. A 62-year-old man with long-standing hypertension and poorly controlled type 2 diabetes mellitus is brought to the emergency department with acute onset confusion and right-sided weakness. On arrival, his blood pressure is 210/120 mmHg. Neurological examination reveals decreased level of consciousness and leftward gaze deviation. CT scan shows no hemorrhage. He is started on an intravenous antihypertensive infusion. Within 30 minutes, his mean arterial pressure (MAP) drops rapidly by 35%. Shortly thereafter, his neurological status worsens significantly, with progression to stupor. Further evaluation suggests cerebral hypoperfusion rather than hemorrhagic transformation. Which of the following best explains the pathophysiological basis for this deterioration?

- a) Failure of metabolic autoregulation due to increased CO₂ causing cerebral vasoconstriction



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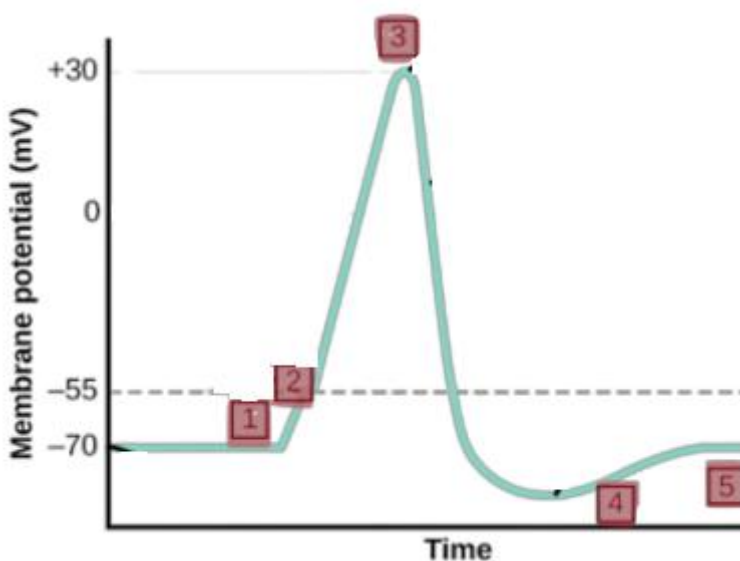
- b) Rightward shift of cerebral autoregulatory curve due to chronic hypertension
- c) Increased intracranial pressure leading to Cushing reflex and reduced cerebral perfusion
- d) Loss of neurovascular coupling due to astrocyte dysfunction

Q24. A 65-year-old male presents with episodes of dizziness and near-syncope over the past few weeks. He has a history of ischemic heart disease. On examination, pulse is irregular with occasional dropped beats. ECG shows regular P waves with constant PR intervals, but intermittently some P waves are not followed by QRS complexes without prior PR prolongation. The abnormality is marked by an arrow in the ECG tracing. Identify the type of heart block seen in this ECG:



- a) First degree block
- b) Mobitz Type 1 block
- c) Mobitz type 2 block
- d) Third degree heart block

Q25. A physiology experiment records the ventricular cardiac action potential using intracellular electrodes. The tracing shows distinct phases labeled from point 1 to 5, representing depolarization, plateau, and repolarization phases. During a specific segment of the graph, the cardiac muscle is completely unresponsive to any external stimulus, preventing tetany and ensuring proper cardiac rhythm. The arrow highlights the region between points 2 and 3, corresponding to the plateau phase of the action potential. Which of the following statements correctly describes this segment?



- a) Point no. 1 to 4 is absolute refractory period
- b) Point no. 2 to 4 is relative refractory period



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- c) Point no. 2 to 3 is absolute refractory period
- d) Point no. 3 to 5 is relative refractory period

- Q26. A 34-year-old male is brought to the clinic after a head injury sustained in a road traffic accident. He is conscious and able to recall past events but is unable to retain any new information given to him after the injury. His language, attention, and motor functions are intact. Neurological evaluation suggests impairment in conversion of short-term memory into long-term memory. Which of the following brain structures is most likely involved in this deficit?
- a) Amygdala
 - b) Hippocampus
 - c) Hypothalamus
 - d) Thalamus
- Q27. A 35-year-old male presents with severe diarrhea for 3 days. He is dehydrated. Arterial blood gas analysis reveals low pH and low bicarbonate levels, consistent with metabolic acidosis. Which of the following compensatory mechanisms is most likely responsible for correcting the acid–base imbalance in this patient?
- a) ↓ Respiration
 - b) ↑ Respiration
 - c) ↓ Renal excretion
 - d) ↑ Bicarbonate loss
- Q28. A 35-year-old man presents with progressive hearing loss. Audiometry suggests a defect in sound transduction at the level of sensory receptors responsible for converting mechanical vibrations into neural signals. Further evaluation points toward dysfunction of specialized hair cells located on the basilar membrane within the inner ear. Which of the following anatomical structures contains these sensory receptors?
- a) Vestibule
 - b) Cochlea
 - c) Semicircular canal
 - d) Tympanic membrane
- Q29. A researcher administers an infusion of insulin to a healthy subject while maintaining euglycemia with simultaneous glucose infusion. Which of the following intracellular changes is most likely to occur in skeletal muscle?
- a) Decreased translocation of GLUT4 to the plasma membrane
 - b) Increased activity of glycogen synthase
 - c) Increased hormone-sensitive lipase activity
 - d) Decreased potassium uptake into cells
- Q30. A patient with diabetic ketoacidosis has the following values: pH 7.21, HCO_3^- 10 mEq/L, PaCO_2 24 mm Hg. Which of the following best describes the acid–base status?
- a) Pure metabolic acidosis with appropriate respiratory compensation
 - b) Metabolic acidosis with superimposed respiratory acidosis
 - c) Metabolic acidosis with superimposed respiratory alkalosis



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d) Pure respiratory acidosis with renal compensation

Q31. A 34-year-old man is brought to high altitude for a research expedition. Within hours, his arterial blood gas shows decreased PaCO_2 , increased pH, and mild reduction in PaO_2 . Over the next 48 hours, his oxygenation improves despite continued low inspired oxygen tension. Which of the following physiological changes most directly explains the early improvement in arterial oxygenation?

- a) Decreased pulmonary vascular resistance in poorly ventilated alveoli
- b) Increased 2,3-BPG shifting the oxyhemoglobin curve to the left
- c) Hypoxic pulmonary vasoconstriction redistributing perfusion
- d) Increased cerebrospinal fluid bicarbonate immediately increasing ventilation

Q32. A patient with severe chronic aortic regurgitation has a markedly elevated stroke volume. Which of the following is most likely to be increased in this patient?

- a) Diastolic arterial pressure
- b) Coronary perfusion pressure during diastole
- c) Pulse pressure
- d) Effective forward cardiac output per beat

Q33. A patient undergoes resection of the terminal ileum. Months later, he develops fat malabsorption despite normal pancreatic function. Which of the following is the most likely underlying physiological defect?

- a) Decreased secretin release causing low bicarbonate secretion
- b) Failure of micelle formation due to bile acid deficiency
- c) Inability to activate pancreatic lipase
- d) Reduced brush-border disaccharidase activity

Q34. A patient with acute hemorrhage has reduced venous return. Which of the following changes would most effectively restore venous return before significant renal compensation occurs?

- a) Decreased sympathetic tone to veins
- b) Increased venous compliance
- c) Increased mean systemic filling pressure
- d) Decreased heart rate to prolong diastole

Q35. A patient with interstitial pulmonary fibrosis has a markedly reduced diffusing capacity for carbon monoxide (DLCO). During exercise, he develops hypoxemia that was not present at rest. Which of the following best explains the worsened hypoxemia during exercise?

- a) Increased mixed venous oxygen content
- b) Reduced alveolar ventilation
- c) Decreased pulmonary capillary transit time
- d) Decreased oxygen consumption by skeletal muscle

Q36. A patient with a pituitary lesion has low ACTH, low cortisol, and preserved aldosterone secretion. Which of the following electrolyte findings is most likely?

- a) Hyperkalemia due to aldosterone deficiency
- b) Hyponatremia due to increased ADH activity



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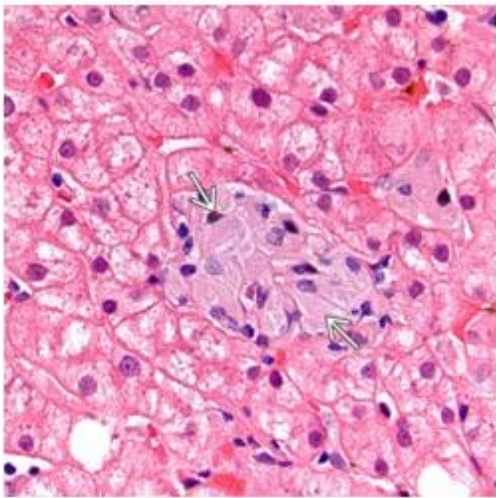
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- c) Hyponatremia due to mineralocorticoid excess
- d) Metabolic alkalosis due to secondary hyperaldosteronism

Q37. A mutation disrupts IRS-1 phosphorylation in skeletal muscle. Which pathway is most directly impaired?

- a) cAMP–PKA signaling
- b) PI3K–Akt pathway
- c) IP₃–DAG pathway
- d) JAK–STAT signaling

Q38. A 10-year-old child presents with progressive abdominal distension, fatigue, and recurrent bone pain. On examination, there is massive splenomegaly and hepatomegaly. Bone marrow biopsy shows macrophages as in the image. Laboratory studies reveal anemia and thrombocytopenia. Which of the following is the most likely underlying enzyme deficiency?



- a) Sphingomyelinase
- b) β -glucocerebrosidase
- c) Hexosaminidase A
- d) Galactocerebrosidase

Q39. A 6-year-old child presents with chronic diarrhea, steatorrhea, and failure to thrive. On examination, the child appears malnourished with evidence of fat-soluble vitamin deficiencies. Laboratory studies reveal markedly reduced levels of chylomicrons in the circulation after a fatty meal. Intestinal biopsy suggests impaired assembly and secretion of chylomicrons from enterocytes into the lymphatic system. The defect is suspected to involve a structural apolipoprotein necessary for chylomicron formation and transport. Which of the following apolipoproteins is essential for chylomicrons to enter the lymphatic system?

- a) Apo B-100
- b) Apo C
- c) Apo E
- d) Apo B-48



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- Q40. A 3-day-old male neonate is brought to the emergency department because of progressive lethargy, poor feeding, tachypnea, and episodic vomiting. He was born at term after an uncomplicated pregnancy and initially appeared well. Physical examination now shows hypotonia and diminished spontaneous movement. Laboratory studies demonstrate severe hyperammonemia, low blood urea nitrogen, elevated plasma glutamine, and increased urinary orotic acid. Arterial blood gas shows respiratory alkalosis. Serum glucose is normal. The infant's mother reports that her brother died in infancy of an undiagnosed neurologic illness. Which of the following is the most likely enzyme deficiency?
- Carbamoyl phosphate synthetase I
 - Ornithine transcarbamylase
 - Argininosuccinate synthetase
 - Arginase
- Q41. A 5-year-old boy is brought in for irritability, abdominal pain, and declining school performance. His family recently moved into an older home undergoing renovation. Examination reveals pallor and mild wrist extensor weakness. Peripheral smear shows microcytic anemia with basophilic stippling. Laboratory testing shows increased free erythrocyte protoporphyrin and elevated aminolevulinic acid. Which of the following enzymes are most directly inhibited in this patient?
- ALA synthase and ferrochelatase
 - ALA dehydratase and ferrochelatase
 - Porphobilinogen deaminase and ferrochelatase
 - Uroporphyrinogen decarboxylase and ferrochelatase
- Q42. 4-month-old infant has recurrent pneumonia, chronic diarrhea, oral thrush, and failure to thrive. Physical examination reveals absent tonsillar tissue and no palpable lymph nodes. Laboratory studies show profound lymphopenia involving both B cells and T cells. Which of the following accumulated metabolites is most directly toxic in this condition?
- dATP
 - ATP
 - cAMP
 - UTP
- Q43. A 7-month-old infant develops repeated vomiting, sweating, tremulousness, and lethargy shortly after fruit puree is introduced into the diet. The child has poor weight gain, jaundice, and hepatomegaly. Reducing substances are absent in the urine when lactose is withheld, but symptoms recur after intake of sucrose-containing foods. Which of the following is the most likely deficient enzyme?
- Fructokinase
 - Aldolase B
 - Galactokinase
 - Hexokinase
- Q44. A tall, thin 12-year-old boy is evaluated for progressive visual problems and recent thromboembolic stroke. Examination reveals downward lens subluxation, mild intellectual disability, and a marfanoid habitus. Laboratory studies show elevated plasma methionine. Which of the following enzymes is most likely deficient?
- Cystathionine β -synthase



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- b) Homogentisate oxidase
- c) Methionine synthase
- d) Phenylalanine hydroxylase

- Q45. A previously healthy 18-month-old child is brought to the emergency department after a viral illness with poor oral intake. The child is lethargic and has a generalized seizure. Laboratory studies show hypoglycemia with low serum ketones, elevated liver enzymes, and increased dicarboxylic acids in the urine. Which of the following is the most likely diagnosis?
- a) Carnitine palmitoyltransferase I deficiency
 - b) Medium-chain acyl-CoA dehydrogenase deficiency
 - c) Pyruvate carboxylase deficiency
 - d) HMG-CoA lyase deficiency
- Q46. A 24-year-old man of Mediterranean ancestry develops dark urine, jaundice, and fatigue two days after being treated with trimethoprim-sulfamethoxazole for a urinary tract infection. Laboratory studies show elevated LDH, decreased haptoglobin, and increased indirect bilirubin. Peripheral smear demonstrates bite cells and Heinz bodies. Deficiency of which of the following most directly predisposed this patient to hemolysis?
- a) Reduced glutathione generation
 - b) Spectrin synthesis
 - c) Heme oxygenase activity
 - d) Pyruvate kinase activity
- Q47. A 7-year-old boy is brought to dermatology because of severe photosensitivity, freckling, and multiple cutaneous malignancies on sun-exposed skin. Neurologic examination is normal. Fibroblasts derived from this patient show inability to repair thymidine dimers induced by ultraviolet radiation. Which of the following DNA repair mechanisms is most likely defective?
- a) Base excision repair
 - b) Nucleotide excision repair
 - c) Mismatch repair
 - d) Nonhomologous end joining
- Q48. A 34-year-old man presents with progressive dyspnea and chronic liver disease. He has no smoking history. CT scan demonstrates panacinar emphysema most prominent at the lung bases. Liver biopsy shows periodic acid-Schiff-positive globules within hepatocytes. Which of the following best explains the hepatic accumulation of the abnormal protein?
- a) Defective ubiquitination of misfolded cytosolic proteins
 - b) Failure of post-translational hydroxylation in the Golgi apparatus
 - c) Misfolding and retention of a mutant secretory protein within the endoplasmic reticulum
 - d) Defective lysosomal targeting due to absent mannose-6-phosphate
- Q49. A 19-year-old man with episodic confusion and vomiting is found to have hyperammonemia due to a partial urea cycle defect. He is treated with a compound that conjugates glycine to form hippurate, thereby facilitating nitrogen excretion. Which of the following agents was most likely administered?
- a) N-carbamylglutamate



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- b) Benzoate
- c) Allopurinol
- d) Rasburicase

- Q50. A 52-year-old man with severe alcohol use disorder is admitted for confusion, gait instability, and ophthalmoplegia. Intravenous glucose is administered before vitamin supplementation, after which his mental status worsens. Deficiency of thiamine pyrophosphate most directly impairs which of the following combinations of enzymes?
- a) Pyruvate dehydrogenase, α -ketoglutarate dehydrogenase, branched-chain α -ketoacid dehydrogenase, transketolase
 - b) Pyruvate carboxylase, citrate synthase, glycogen phosphorylase, transaminase
 - c) Carbamoyl phosphate synthetase I, pyruvate dehydrogenase, succinate dehydrogenase, transketolase
 - d) Ornithine transcarbamylase, α -ketoglutarate dehydrogenase, transaminase, succinyl-CoA synthetase
- Q51. A 9-year-old child is born with a mutation causing failure of autoreactive lymphocytes to undergo programmed cell death during development. The affected protein normally interacts with Fas-associated death domain protein (FADD) after receptor ligation. Which of the following enzymes is most directly activated downstream of this pathway?
- a) Telomerase
 - b) Caspase-8
 - c) Cyclin-dependent kinase 4
 - d) Phospholipase A2
- Q52. A 55-year-old patient presents with progressive fatigue, weight loss, and signs of organ dysfunction. A biopsy from the affected tissue shows extracellular deposition of amorphous eosinophilic material. Special staining is performed, and under polarized light, the deposits exhibit apple-green birefringence. Which of the following stains is most likely responsible for this finding?
- a) H&E
 - b) PAS
 - c) Congo red
 - d) Gram stain
- Q53. A 32-year-old woman presents with cough, dyspnea, and bilateral hilar lymphadenopathy. Biopsy of a mediastinal lymph node shows tightly clustered epithelioid histiocytes and multinucleated giant cells without central necrosis. Which cytokine is most important for maintenance of this type of inflammatory response?
- a) IL-4
 - b) IL-5
 - c) IFN- γ
 - d) IL-10
- Q54. A healthy 19-year-old man undergoes clean surgical excision of a superficial skin lesion with immediate approximation of the wound edges. Which of the following is most characteristic of healing by primary intention?
- a) Large tissue defect with abundant granulation tissue and significant wound contraction



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- b) Minimal scarring because epithelial regeneration occurs without granulation tissue formation
- c) Small amount of granulation tissue formation with relatively limited collagen deposition
- d) Dense neutrophilic abscess formation as a required intermediate step

Q55. A 6-year-old boy presents with periorbital swelling and generalized edema. Urinalysis reveals 4+ proteinuria with oval fat bodies. Serum albumin is markedly reduced. Which of the following is the principal mechanism contributing to his edema?

- a) Increased plasma oncotic pressure due to retention of albumin in the vascular space
- b) Increased hydrostatic pressure caused by left-sided heart failure
- c) Decreased plasma oncotic pressure due to urinary loss of albumin
- d) Lymphatic obstruction due to parasitic infection

Q56. A 72-year-old man with long-standing hypertension and ischemic cardiomyopathy presents with progressive dyspnea and orthopnea. Lung biopsy, performed for an unrelated reason, shows numerous macrophages containing coarse golden-brown cytoplasmic pigment in alveolar spaces. Prussian blue stain is positive. Which of the following is the most likely source of this pigment?

- a) Lipofuscin from free radical injury to macrophage membranes
- b) Bilirubin from breakdown of alveolar surfactant
- c) Hemosiderin derived from phagocytosed red blood cells after chronic pulmonary congestion
- d) Anthracotic carbon from cigarette smoke accumulation

Q57. A 64-year-old woman with severe right-sided heart failure dies of progressive multiorgan dysfunction. On autopsy, the liver has a mottled red-and-tan cut surface often described as “nutmeg liver.” Which of the following histologic changes is most likely present?

- a) Diffuse fatty change centered in periportal hepatocytes only
- b) Centrilobular sinusoidal congestion with hepatocyte necrosis near central veins
- c) Noncaseating granulomas throughout portal tracts
- d) Massive neutrophilic infiltration of bile ducts

Q58. A 46-year-old woman undergoes cervical biopsy after an abnormal screening test. Histology shows full-thickness epithelial atypia with hyperchromatic pleomorphic nuclei, loss of maturation, and numerous mitotic figures, but the basement membrane remains intact. Which of the following is the most accurate diagnosis?

- a) Mild dysplasia
- b) Metaplasia
- c) Carcinoma in situ
- d) Invasive squamous cell carcinoma

Q59. A child develops bilateral retinoblastomas at a very young age. Genetic testing reveals one germline mutation and one acquired somatic mutation affecting the same tumor suppressor gene. Which of the following best explains this pattern?

- a) Dominant negative oncogene activation requiring only one mutational event
- b) Two-hit inactivation of both alleles of a tumor suppressor gene
- c) Trinucleotide repeat expansion causing genomic imprinting



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d) Mitochondrial inheritance of a growth factor receptor mutation

Q60. A 59-year-old heavy smoker presents with cough, weight loss, proximal muscle weakness, new-onset hypertension, and hyperglycemia. Chest imaging shows a centrally located hilar mass. Laboratory studies reveal elevated ACTH and cortisol levels. Which of the following tumors is most likely responsible?

- a) Lung adenocarcinoma
- b) Small cell carcinoma of the lung
- c) Squamous cell carcinoma of the lung
- d) Carcinoid tumor of the lung

Q61. A 12-year-old boy develops cola-colored urine, periorbital edema, and hypertension two weeks after recovering from streptococcal pharyngitis. Urinalysis reveals red blood cell casts and mild proteinuria. Which of the following is the most likely pathogenic mechanism?

- a) Autoantibodies directed against podocyte phospholipase A2 receptor
- b) Immune complex deposition with complement activation in glomeruli
- c) Linear antibody deposition against type IV collagen
- d) Nonenzymatic glycosylation of glomerular basement membrane proteins

Q62. A 48-year-old man with severe pancreatitis develops rapidly progressive respiratory failure requiring mechanical ventilation. Chest imaging shows diffuse bilateral pulmonary infiltrates. Histology reveals alveolar septal edema and eosinophilic hyaline membranes lining alveolar spaces. Which of the following best describes the underlying pulmonary process?

- a) Chronic bronchitis due to mucus gland hyperplasia
- b) Diffuse alveolar damage with increased alveolar-capillary permeability
- c) Cardiogenic pulmonary edema due to elevated left atrial pressure alone
- d) Caseating granulomatous inflammation involving terminal bronchioles

Q63. A 27-year-old man presents with malaise, hemoptysis, facial puffiness, and dark urine for 5 days. Two weeks earlier he had mild myalgias but no sore throat or skin infection. Blood pressure is 168/102 mm Hg. Laboratory studies show elevated creatinine, dysmorphic red blood cells in urine, red blood cell casts, and mild proteinuria. Chest imaging demonstrates patchy bilateral alveolar opacities. Renal biopsy shows crescent formation in most glomeruli. Immunofluorescence demonstrates smooth, linear staining of the glomerular basement membrane with IgG and C3. Which of the following is the most likely underlying mechanism?

- a) Deposition of circulating immune complexes after streptococcal infection
- b) Autoantibodies directed against noncollagenous domains of type IV collagen
- c) Pauci-immune small-vessel vasculitis associated with PR3-ANCA
- d) Mesangial IgA deposition after mucosal infection

Q64. 64-year-old man with longstanding exertional angina is prescribed a transdermal nitroglycerin patch. During follow-up, he reports that the medication worked very well during the first week, but after continuous daily use, he now experiences much less symptomatic relief despite proper adherence. His physician modifies the prescription so that the patch is removed each night for several hours before a new one is applied the next morning. Which of the following best explains the rationale for this change?



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- a) Prevention of reflex tachycardia caused by chronic β_1 -receptor upregulation
 - b) Avoidance of depletion of sulfhydryl groups required for nitric oxide generation
 - c) Reduction of irreversible guanylyl cyclase desensitization in vascular smooth muscle
 - d) Prevention of increased hepatic first-pass metabolism of nitrates
- Q65. 72-year-old man with ischemic heart disease and prior myocardial infarction develops sustained monomorphic ventricular tachycardia. He is started on an antiarrhythmic agent that blocks inactivated sodium channels and also prolongs the action potential duration. The drug is particularly effective in ischemic ventricular tissue but is associated with torsades de pointes. Which of the following drugs was most likely administered?
- a) Lidocaine
 - b) Quinidine
 - c) Flecainide
 - d) Procainamide
- Q66. A 59-year-old man with diabetic nephropathy and hypertension is started on lisinopril. Two weeks later, his blood pressure is improved, but laboratory studies reveal a modest rise in serum creatinine and mild hyperkalemia. Which of the following renal hemodynamic changes most directly explains the increase in serum creatinine?
- a) Constriction of the afferent arteriole with reduction in renal plasma flow
 - b) Dilation of the efferent arteriole with reduction in intraglomerular pressure
 - c) Enhanced sodium delivery to the macula densa with afferent vasodilation
 - d) Increased mesangial cell contraction leading to reduced filtration surface area
- Q67. A 68-year-old woman with decompensated systolic heart failure is admitted for pulmonary edema. She receives an intravenous medication that rapidly improves dyspnea by inhibiting sodium-potassium-2 chloride cotransport in the thick ascending limb of the loop of Henle. Which of the following additional effects is most likely to occur with this therapy?
- a) Increased calcium reabsorption due to enhanced luminal positivity
 - b) Decreased uric acid excretion with risk of gout
 - c) Metabolic acidosis due to reduced hydrogen secretion
 - d) Decreased prostaglandin synthesis in the kidney
- Q68. A 19-year-old college student with persistent asthma continues to have nighttime awakenings and frequent use of a rescue inhaler despite appropriate technique. A controller medication is added that binds intracellular receptors, alters gene transcription, and decreases eosinophilic airway inflammation. Which of the following additional molecular effects is most directly responsible for the therapeutic benefit of this drug class?
- a) Direct competitive antagonism of leukotriene D₄ at the CysLT₁ receptor
 - b) Increased histone acetylation at proinflammatory cytokine loci
 - c) Decreased transcription of inflammatory mediators through inhibition of NF- κ B activity
 - d) Activation of membrane guanylyl cyclase in bronchial smooth muscle
- Q69. A 14-year-old boy has asthma symptoms that are especially prominent after exercise and after accidental exposure to aspirin-containing medications. He is prescribed a drug that prevents leukotriene-



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mediated bronchoconstriction without inhibiting cyclooxygenase. Which of the following medications most likely was prescribed?

- a) Zileuton
- b) Montelukast
- c) Theophylline
- d) Omalizumab

Q70. A 48-year-old man with chronic gastroesophageal reflux disease is prescribed omeprazole. He is told to take the medication before meals rather than afterward. Which of the following best explains the importance of this timing?

- a) The drug requires bile salts for intestinal absorption
- b) It irreversibly inhibits H^+/K^+ ATPase only after activation in acidic canaliculi of parietal cells
- c) It must be taken with food to reduce rapid hepatic metabolism
- d) It prevents gastrin release only when administered during fasting

Q71. A 29-year-old woman with moderate to severe Crohn disease has persistent symptoms despite corticosteroids and azathioprine. She is started on a monoclonal antibody that neutralizes $TNF-\alpha$. Before treatment, screening reveals latent tuberculosis, and therapy is delayed until prophylaxis is initiated. Which of the following drugs was most likely planned?

- a) Mesalamine
- b) Infliximab
- c) Loperamide
- d) Ondansetron

Q72. A 26-year-old man with schizophrenia is started on haloperidol for severe agitation and auditory hallucinations. Several days later, he develops neck stiffness, jaw spasm, and upward deviation of the eyes. Which of the following best explains the pathophysiology of this adverse effect?

- a) Increased serotonergic tone in the raphe nuclei
- b) Muscarinic receptor hypersensitivity in the striatum
- c) Dopamine D2 receptor blockade in the nigrostriatal pathway
- d) Irreversible destruction of dopaminergic neurons in the substantia nigra

Q73. A 33-year-old woman taking sertraline for depression begins linezolid for resistant pneumonia. Within 24 hours she develops agitation, diaphoresis, diarrhea, hyperreflexia, clonus, and fever. Which of the following best describes the most likely pharmacologic basis of her condition?

- a) Excess central dopamine blockade causing malignant hyperthermia-like toxicity
- b) Synergistic enhancement of serotonergic neurotransmission
- c) Sudden GABA withdrawal after abrupt benzodiazepine cessation
- d) Irreversible NMDA receptor antagonism in the cortex

Q74. A 22-year-old woman with focal seizures is started on carbamazepine. Several weeks later, her seizure frequency decreases, but she develops dizziness, diplopia, and mild hyponatremia. Which of the following best describes the primary antiseizure mechanism of this drug?

- a) Enhancement of GABA-A receptor chloride current duration
- b) Inhibition of T-type calcium channels in thalamic neurons



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- c) Stabilization of the inactivated state of voltage-gated sodium channels
- d) Irreversible inhibition of GABA transaminase

Q75. A 49-year-old woman with a deep venous thrombosis is started on warfarin without heparin bridging. Three days later, she develops painful, erythematous skin lesions that become necrotic. Which of the following best explains this complication?

- a) Rapid decline in protein C levels before full reduction of procoagulant clotting factors
- b) Immune-mediated platelet activation caused by antibodies to platelet factor 4
- c) Suppression of vitamin K absorption due to intestinal dysbiosis
- d) Excess plasmin generation causing dermal hemorrhage

Q76. 42-year-old woman with episodic headaches, diaphoresis, palpitations, and paroxysmal hypertension is diagnosed with pheochromocytoma. Before adrenalectomy, she is given a nonselective, irreversible α -adrenergic antagonist, followed later by a β -blocker. Why must the α -blocker be given first?

- a) β -blockade first would precipitate severe hypotension from unopposed muscarinic stimulation
- b) β -blockade first would prevent reflex bradycardia needed for hemodynamic stability
- c) β -blockade first would leave unopposed α -adrenergic vasoconstriction and worsen hypertension
- d) α -blockade first is required to prevent acetylcholine release from postganglionic fibers

Q77. A 72-year-old woman is hospitalized for aspiration pneumonia and receives broad-spectrum antibiotics. Five days later, she develops fever, leukocytosis, abdominal cramping, and profuse watery diarrhea. CT scan demonstrates diffuse colonic wall thickening. Colonoscopy reveals raised yellow-white plaques overlying erythematous mucosa. The causative organism produces toxins that most directly exert their pathogenic effect through which of the following mechanisms?

- a) ADP-ribosylation of Gs protein causing increased cAMP
- b) Inactivation of the 60S ribosomal subunit by removing adenine from rRNA
- c) Glucosylation of Rho GTPases with disruption of the actin cytoskeleton
- d) Cleavage of synaptobrevin causing inhibition of acetylcholine release

Q78. A 43-year-old man sustains a deep crush injury to his calf in a farming accident. Twelve hours later, he develops rapidly progressive pain out of proportion to exam, edema, hemorrhagic bullae, crepitus, and hypotension. Surgical exploration reveals devitalized muscle with a thin brownish exudate. Gram stain shows large gram-positive rods with few leukocytes. Which of the following virulence factors is most responsible for the extensive myonecrosis?

- a) Lecithinase that degrades cell membranes
- b) Heat-stable enterotoxin acting as a superantigen
- c) Metalloprotease that cleaves SNARE proteins
- d) M protein that inhibits opsonization

Q79. A 5-year-old child presents with paroxysmal coughing fits followed by post-tussive emesis. Nasopharyngeal culture grows a fastidious gram-negative coccobacillus. The organism undergoes phase variation affecting virulence gene expression. Which of the following virulence factors is most responsible for lymphocytosis in this infection?

- a) Tracheal cytotoxin
- b) Adenylate cyclase toxin



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- c) Pertussis toxin
- d) Filamentous hemagglutinin

Q80. A 45-year-old man from the Mississippi River valley develops chronic cough, weight loss, and hepatosplenomegaly. Biopsy reveals granulomas with yeast forms inside macrophages. Which virulence factor allows intracellular survival?

- a) Melanin production
- b) Capsule formation
- c) Inhibition of phagolysosome acidification
- d) Dimorphic transition regulated by temperature

Q81. A 2-day-old newborn develops poor feeding, lethargy, temperature instability, and episodes of apnea. Blood cultures are pending. Lumbar puncture demonstrates neutrophilic pleocytosis, elevated protein, and low glucose. The mother received inconsistent prenatal care and was not screened late in pregnancy. Vaginal colonization by which of the following organisms is the most likely source of this infection?

- a) *Listeria monocytogenes*
- b) *Streptococcus agalactiae*
- c) *Escherichia coli* O157:H7
- d) *Neisseria gonorrhoeae*

Q82. A pregnant woman at 30 weeks' gestation develops fever, myalgias, and diarrhea two days after eating unpasteurized soft cheese. Blood cultures later grow a motile gram-positive rod. The organism invades host cells, escapes the phagosome, and spreads directly from cell to cell without major extracellular exposure. Which virulence factor most directly permits escape from the phagosome?

- a) Lecithinase
- b) Listeriolysin O
- c) M protein
- d) Endotoxin lipid A

Q83. A 52-year-old man who has not sought medical care in decades presents with painful muscle stiffness and difficulty opening his mouth 10 days after stepping on a rusty nail while working outdoors. Examination reveals risus sardonicus, trismus, and episodic painful spasms triggered by minimal stimulation. Which of the following best describes the mechanism of the causative toxin?

- a) Inactivation of the 60S ribosomal subunit in enterocytes
- b) Blockade of glycine and GABA release from inhibitory interneurons
- c) Activation of guanylyl cyclase in intestinal epithelial cells
- d) ADP-ribosylation of EF-2 in cardiac myocytes

Q84. A 17-year-old boy presents with fever, severe headache, photophobia, and a rapidly progressive petechial rash. He is hypotensive and confused. Blood cultures grow oxidase-positive gram-negative diplococci. Review of records reveals two prior episodes of invasive infection with the same organism. Which of the following immune defects most likely explains this pattern?

- a) Deficiency of C5-C9 terminal complement components
- b) Deficiency of NADPH oxidase in neutrophils



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- c) Deficiency of IL-12 receptor signaling
- d) Selective IgA deficiency

- Q85. A 22-year-old man presents with dysuria and purulent urethral discharge 4 days after unprotected intercourse. Gram stain of the discharge shows numerous neutrophils containing intracellular gram-negative diplococci. Which of the following is the most important virulence mechanism that allows the causative organism to evade host immunity and cause recurrent infection?
- a) Antigenic variation of pili and outer membrane proteins
 - b) Formation of latent episomal DNA in epithelial cells
 - c) Release of a neurotoxin that blocks acetylcholine release
 - d) Survival within macrophages by preventing phagolysosome fusion
- Q86. A 63-year-old man with extensive burns develops fever, hypotension, and blue-green purulent drainage from the wound. Blood cultures grow a non-lactose-fermenting gram-negative rod that produces a fruity odor. One of the organism's major toxins exerts its effect by which mechanism?
- a) Activation of guanylyl cyclase causing increased cGMP
 - b) ADP-ribosylation of elongation factor-2
 - c) Inactivation of the 60S ribosome by depurination
 - d) Cleavage of SNARE proteins in motor neurons
- Q87. A 67-year-old man with alcoholism presents with fever, productive cough, pleuritic chest pain, and rust-colored sputum. Chest radiograph shows lobar consolidation. Sputum Gram stain shows lancet-shaped gram-positive diplococci. Which of the following is the most important virulence factor that permits this organism to evade phagocytosis and cause invasive disease?
- a) M protein
 - b) Polysaccharide capsule
 - c) Catalase
 - d) Coagulase
- Q88. A 45-year-old man develops severe vomiting, abdominal pain, watery diarrhea, hypotension, and later peripheral neuropathy after chronic exposure to contaminated groundwater. On examination, there is hyperpigmentation of the skin and hyperkeratosis of the palms and soles. Which of the following poisons most likely explains this constellation of findings?
- a) Lead
 - b) Arsenic
 - c) Mercury
 - d) Carbon monoxide
- Q89. A 38-year-old man consumes an illicit alcohol preparation during a social gathering. Within hours he develops vomiting, headache, blurred vision, and progressive confusion. Arterial blood gas shows a high anion gap metabolic acidosis. Which of the following toxic metabolites is most responsible for his visual symptoms?
- a) Acetaldehyde
 - b) Formic acid
 - c) Oxalic acid



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d) Acetic acid

Q90. A young man is found unconscious in a public restroom with slow breathing and pinpoint pupils. Needle marks are noted in the antecubital fossa. Which of the following findings is most consistent with opioid poisoning?

- a) Hyperthermia, mydriasis, and severe agitation
- b) Coma, respiratory depression, and miosis
- c) Dry skin, urinary retention, and delirium
- d) Blindness with metabolic acidosis

Q91. A charred body is found in a warehouse fire. There are large split-like defects over the limbs and trunk. The margins are irregular, but there is no vital reaction, no surrounding bruising, and underlying vessels are intact. Which of the following best describes these lesions?

- a) Incised wounds
- b) Antemortem lacerations caused by blunt force trauma
- c) Heat ruptures
- d) Chop wounds

Q92. In a disputed juvenile case, an X-ray of the elbow is used for age estimation. Which of the following regions is classically useful because multiple secondary ossification centers appear there in a predictable sequence?

- a) Shoulder girdle only
- b) Elbow joint region
- c) Skull vault only
- d) Sternum only

Q93. In a discussion of execution methods, a medical student asks why properly performed judicial hanging often causes rapid death. Which of the following is the classic fatal injury in such cases?

- a) Fracture-dislocation of upper cervical vertebrae
- b) Bilateral femoral artery rupture
- c) Immediate rupture of the thoracic aorta
- d) Pulmonary embolism from lower limb venous thrombosis

Q94. A child is brought dead with multiple bluish-purple marks over the buttocks and posterior thighs. Investigators want to distinguish inflicted blunt trauma from postmortem lividity. Which of the following findings most strongly supports that these lesions are true antemortem bruises?

- a) The discoloration is located on dependent parts of the body
- b) Incision into the area shows infiltration of blood into tissues
- c) The color is purple-blue
- d) The body was found supine

Q95. A 33-year-old construction worker is crushed under a heavy concrete slab across the chest for several minutes before rescue. He dies shortly after arrival. On examination there are intense petechial hemorrhages over the face, neck, and conjunctivae with cyanosis and upper chest congestion. Which of the following best explains this pattern?



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- a) Compression of the thorax preventing venous return from the head and neck
 - b) Selective rupture of carotid arteries by hyperextension
 - c) Aspiration of blood into alveoli causing diffuse hypoxia
 - d) Reflex cardiac arrest from vagal stimulation by neck pressure
- Q96. A 19-year-old woman presents to the emergency department alleging sexual assault that occurred several hours earlier. She is physically stable. Which of the following is the most appropriate immediate medicolegal approach?
- a) Refuse examination until police arrive with a written court order
 - b) Obtain informed consent, provide medical care, document findings carefully, and preserve forensic evidence
 - c) Prioritize evidence collection only and delay treatment until the examination is complete
 - d) Examine only genital injuries because non-genital injuries are irrelevant
- Q97. A 49-year-old woman undergoes total thyroidectomy for papillary thyroid carcinoma. On the first postoperative day, she has a weak, breathy voice and frequent coughing whenever she drinks water. Laryngoscopy shows the right true vocal fold fixed in a paramedian position. Which of the following nerves was most likely damaged?
- a) External branch of superior laryngeal nerve
 - b) Internal branch of superior laryngeal nerve
 - c) Recurrent laryngeal nerve
 - d) Auricular branch of vagus nerve
- Q98. A 4-year-old unimmunized boy is brought to the emergency department with high fever, severe sore throat, muffled voice, drooling, and inspiratory stridor. He is sitting upright, leaning forward, and refuses to lie down. He appears anxious and toxic. Lateral neck radiograph shows an enlarged, epiglottic shadow. Which of the following is the most immediate priority in management?
- a) Perform throat examination with a tongue depressor to visualize the lesion
 - b) Secure the airway in a controlled setting
 - c) Discharge with oral antibiotics and close follow-up
 - d) Order CT scan of the neck before intervention
- Q99. A 28-year-old man presents with progressive unilateral hearing loss. Weber testing lateralizes to the right ear. On Rinne testing, bone conduction is greater than air conduction on the right, whereas air conduction remains greater than bone conduction on the left. Which of the following best explains these findings?
- a) Right sensorineural hearing loss
 - b) Left conductive hearing loss
 - c) Right conductive hearing loss
 - d) Left sensorineural hearing loss
- Q100. A 34-year-old woman reports gradually progressive bilateral hearing loss, worse during pregnancy. Otoscopic examination is normal. Weber test lateralizes to the more affected ear, and Rinne is negative on that side. Which of the following is the most likely underlying pathology?
- a) Destruction of the organ of Corti



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- b) Fixation of the stapes footplate at the oval window
- c) Recurrent serous otitis media due to eustachian tube dysfunction
- d) Acoustic neuroma compressing CN VIII

Q101. A 56-year-old woman complains of slowly progressive unilateral hearing loss, tinnitus, and imbalance. MRI reveals a cerebellopontine angle mass arising from the vestibular portion of CN VIII. If the mass enlarges further, which of the following additional findings is most likely to develop next?

- a) Contralateral homonymous hemianopia
- b) Ipsilateral facial weakness
- c) Loss of gag reflex only
- d) Bilateral abducens palsy without other deficits

Q102. A patient presents with complete right facial paralysis. He also complains that ordinary sounds are painfully loud, food on the anterior tongue has lost taste, and the right eye feels abnormally dry. Which of the following is the most likely site of the lesion?

- a) Distal to the stylomastoid foramen
- b) Within the parotid gland, distal to all major branches
- c) Proximal to the geniculate ganglion within the facial canal
- d) Isolated lesion of the chorda tympani

Q103. A 30-year-old man is involved in a motor vehicle collision and sustains anterior skull base trauma. He later develops clear unilateral nasal discharge that increases when leaning forward. Laboratory testing confirms β 2-transferrin in the fluid. Which of the following structures was most likely fractured?

- a) Lamina papyracea
- b) Cribriform plate
- c) Hard palate
- d) Maxillary sinus floor

Q104. A 7-year-old child with acute sinusitis develops painful eye movements, fever, eyelid swelling, and proptosis. Which of the following anatomic features best explains the spread of infection from the sinus into the orbit?

- a) Direct continuity through the optic canal
- b) Thin lamina papyracea separating ethmoid sinus from orbit
- c) Connection through the eustachian tube
- d) Drainage of the maxillary sinus into the inferior meatus

Q105. A 44-year-old diabetic man develops rapid swelling of the submandibular region after an infected lower molar tooth. He has fever, drooling, dysphagia, and elevation of the tongue. The floor of mouth is tense and indurated. Which of the following spaces is primarily involved?

- a) Retropharyngeal space
- b) Submandibular and sublingual spaces
- c) Parotid space
- d) Carotid sheath



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- Q106. A 42-year-old woman reports episodic severe vertigo lasting hours, fluctuating low-frequency hearing loss, tinnitus, and aural fullness in one ear. Neurologic examination between attacks is normal. Which of the following is the most likely underlying pathology?
- a) Acute bacterial labyrinthitis
 - b) Endolymphatic hydrops
 - c) Fixed stapes footplate
 - d) Vestibular schwannoma
- Q107. A 29-year-old woman presents with subacute visual blurring in her right eye over 3 days. She reports pain that worsens with eye movement and difficulty distinguishing colors, especially red. Visual acuity is reduced in the affected eye. Swinging flashlight test demonstrates dilation of both pupils when the light is moved from the left eye to the right eye. Funduscopic examination is unremarkable. MRI of the brain later reveals several periventricular white matter lesions. Which of the following is the most likely site of the primary lesion causing her visual symptoms?
- a) Macula of the right retina
 - b) Right optic nerve anterior to the chiasm
 - c) Right lateral geniculate nucleus
 - d) Left optic tract
- Q108. A 72-year-old man with atrial fibrillation develops sudden difficulty seeing the left half of the visual world. Neurologic examination is otherwise unremarkable. Confrontation testing shows a left homonymous hemianopia with preservation of central vision. Which vascular lesion is most likely responsible?
- a) Right ophthalmic artery occlusion
 - b) Left central retinal artery occlusion
 - c) Right posterior cerebral artery infarction
 - d) Right middle cerebral artery superior division infarction
- Q109. A 68-year-old man with hypertension and carotid atherosclerosis develops sudden, profound painless loss of vision in his right eye while reading the newspaper. Funduscopic examination shows diffuse retinal pallor with a prominent cherry-red spot at the fovea. Which of the following best explains the foveal appearance?
- a) The fovea is supplied mainly by the central retinal artery, so it becomes hyperemic first
 - b) The fovea lacks a retinal blood supply and its underlying choroidal circulation remains visible
 - c) The optic disc becomes infarcted before the fovea
 - d) Vitreous hemorrhage selectively spares the fovea
- Q110. A 63-year-old woman is brought to the emergency department with sudden severe right eye pain, headache, nausea, and blurred vision with colored halos around lights. The eye is red, the cornea appears hazy, and the pupil is mid-dilated and poorly reactive. Which of the following is the most likely underlying mechanism?
- a) Autoimmune destruction of lacrimal glands causing dry corneal surface
 - b) Obstruction of trabecular meshwork outflow by closure of the iridocorneal angle
 - c) Progressive open-angle degeneration of optic nerve without rise in intraocular pressure



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d) Retinal detachment from vitreoretinal traction

Q111. A 61-year-old man with longstanding diabetes is found on screening to have elevated intraocular pressure and increased cup-to-disc ratio bilaterally. He denies pain. Visual field testing later demonstrates gradual peripheral field constriction. Which pattern of visual loss is most typical of advanced primary open-angle glaucoma?

- a) Central scotoma only
- b) Bitemporal hemianopia
- c) Tunnel vision
- d) Homonymous hemianopia

Q112. A 27-year-old woman is found to have anisocoria during a routine examination. The right pupil is larger than the left and reacts poorly to light, but constricts slowly with near effort and redilates slowly. Deep tendon reflexes are diminished. Which of the following is the most likely diagnosis?

- a) Argyll Robertson pupil
- b) Horner syndrome
- c) Adie tonic pupil
- d) Oculomotor nerve compression

Q113. A 56-year-old man with gait instability and impaired vibration sense in the lower limbs is noted to have small irregular pupils that constrict during accommodation but fail to constrict in response to light. Which of the following is the most likely underlying diagnosis?

- a) Neurosyphilis
- b) Multiple sclerosis
- c) Diabetic retinopathy
- d) Acute glaucoma

Q114. A 33-year-old man presents with severe photophobia, eye pain, and blurred vision. Examination reveals ciliary flush, consensual photophobia, and a small irregular pupil. Slit-lamp exam shows cells and flare in the anterior chamber. Which of the following is the most likely diagnosis?

- a) Acute conjunctivitis
- b) Anterior uveitis
- c) Open-angle glaucoma
- d) Vitreous detachment

Q115. A patient with long-standing diabetes reports progressive painless decline in vision and increasing glare while driving at night. Ophthalmoscopy shows loss of lens transparency. Which structure has become pathologically opaque?

- a) Corneal stroma
- b) Vitreous body
- c) Crystalline lens
- d) Aqueous humor

Q116. A newborn is found to have a keyhole-shaped defect in the inferior iris. Which embryologic event most likely failed to occur properly?



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- a) Fusion of the optic cup with surface ectoderm
- b) Closure of the embryonic choroidal fissure
- c) Canalization of the nasolacrimal duct
- d) Regression of the primary vitreous only

Q117. A 46-year-old woman undergoes total thyroidectomy for multinodular goiter causing compressive symptoms. Six hours after surgery, she develops progressive neck tightness, dysphagia, muffled voice, and increasing respiratory distress. Examination reveals a tense swelling beneath the incision, stridor, and agitation. Oxygen saturation is falling despite supplemental oxygen. The junior resident considers obtaining an urgent neck ultrasound before intervention. Which of the following is the most appropriate immediate next step?

- a) Administer intravenous dexamethasone and observe closely
- b) Obtain an urgent CT scan of the neck to define the collection
- c) Open the wound sutures at the bedside and evacuate the hematoma
- d) Reintubate first and then return to the operating room later

Q118. A 63-year-old man undergoes elective sigmoid colectomy for recurrent diverticulitis. On postoperative day 1, he is febrile to 38.3°C (100.9°F). He has mild incisional discomfort but no peritoneal signs. Lungs are clear, oxygen saturation is normal, urine output is adequate, and the surgical wound is clean. Which of the following is the most likely cause of his fever?

- a) Anastomotic leak
- b) Atelectasis-induced inflammatory response
- c) Deep surgical site infection
- d) Pulmonary embolism

Q119. A 24-year-old man is brought after a motorcycle collision. CT scan reveals a grade III splenic laceration with moderate hemoperitoneum but no active contrast extravasation. Blood pressure is 118/72 mm Hg, pulse 96/min, and he responds appropriately to questions. Serial hemoglobin levels are stable over 6 hours. Which of the following is the most appropriate management?

- a) Immediate exploratory laparotomy and splenectomy
- b) Nonoperative management with close monitoring
- c) Emergency distal pancreatectomy with splenectomy
- d) Diagnostic peritoneal lavage

Q120. A 44-year-old obese woman presents with recurrent episodes of right upper quadrant pain after fatty meals. On this occasion, the pain has lasted 14 hours and is associated with nausea and low-grade fever. Examination reveals localized right upper quadrant tenderness and inspiratory arrest on palpation beneath the costal margin. Leukocyte count is elevated. Ultrasound shows gallstones, gallbladder wall thickening, and pericholecystic fluid. Which of the following is the most likely diagnosis?

- a) Choledocholithiasis
- b) Acute calculous cholecystitis
- c) Ascending cholangitis
- d) Biliary colic without inflammation



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Q121. A 45-year-old man attends a health camp under given program. His blood pressure is 130/86 mmHg. According to national guidelines, screening for hypertension under program begins at what age?



- a) 18 years- home
- b) 30years- PHC
- c) 30 years- sub centre
- d) 40 years- sub centre

Q122. A 58-year-old daily wage worker requires coronary artery bypass surgery. He is enrolled under given program, and is eligible for cashless hospitalization. What is the annual coverage amount under program



- a) ₹2 lac per person/year
- b) ₹2 lakh / family/year
- c) ₹5 lakh/ person/year
- d) ₹5 lakh / family/year

Q123. A 26-year-old pregnant woman tests positive for HIV during routine antenatal screening. She is asymptomatic. As per national guidelines, what is the recommended strategy to prevent mother-to-child transmission?

- a) Start ART only in third trimester to mother
- b) Single dose Nevirapine during labour to mother and newborn
- c) Triple-drug ART to mother throughout pregnancy irrespective of CD4 count
- d) Triple-drug ART to mother throughout pregnancy irrespective of CD4 count with newborn prophylaxis with nevirapine

Q124. A 24-year-old migrant worker presents with high fever, chills, and splenomegaly with blackish skin discolouration. Urine RK39 test positive. According to national guidelines, all are Incorrect about this disease

- a) Disease is not included under national vector born disease control program
- b) Anopheles mosquito is vector
- c) DDT spray for vector control up to 6feet height from ground level



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d) ASHA is conducting monthly fortnight survey

Q125. A 6-week-old infant is brought to the PHC for routine immunization. The ANM administers Pentavalent vaccine, OPV-1, Rotavirus vaccine, PCV 1, and IPV (fractional dose). The fractional IPV fl is administered at which anatomical site?

- Intramuscular in anterolateral thigh Lt
- Subcutaneous in LT arm
- Intradermal in right upper arm
- Intradermal in left upper arm

Q126. A 32-year-old male presents to a government health centre with chronic cough for 3 weeks, evening rise of temperature, and weight loss. Sputum CBNAAT confirms rifampicin-sensitive pulmonary tuberculosis. He is registered under the national TB program. The Medical Officer informs him that he will receive a Direct Benefit Transfer (DBT) for nutritional support during treatment. Under the current National TB Elimination Programme (NTEP), what is the amount provided for nutritional support?

- ₹500/ month till end of treatment
- ₹1000/ month till end of treatment
- ₹1000/ one time
- ₹1000/ month for 6 months irrespective of type of TB

Q127. A 60-year-old man suffers a stroke and lives with disability for 5 years before dying 10 years prematurely. The measure combining years of life lost and years lived with disability is:

- QALY
- DALY
- Incidence rate
- Case fatality rate

Q128. A 42-year-old asymptomatic woman undergoes VIA (Visual Inspection with Acetic Acid) at a PHC under NPCDCS. Which level of prevention does this represent?



- Primordial prevention
- Primary prevention
- Secondary prevention
- Tertiary prevention

Q129. A 35-year-old woman has:

- Weight = 78 kg
- Height = 1.6 m

BMI = ?



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- a) Overweight
- b) Obesity
- c) Underweight
- d) Normal

Q130. In a rural population, 12% individuals are known diabetic. On active screening, an additional 18% are newly diagnosed. Which epidemiological concept is demonstrated?

- a) Secular trend
- b) Iceberg phenomenon
- c) Surveillance bias
- d) Screening paradox

Q131. A 3-year-old child presents with barking cough, inspiratory stridor, and hoarseness. Several similar cases are reported in the community. The disease spreads mainly by droplet infection. What level of prevention would mass vaccination against this disease represent?

- a) Primordial prevention
- b) Primary prevention
- c) Secondary prevention
- d) Tertiary prevention

Q132. A district reports 200 new tuberculosis cases in a population of 2 lakh during one year. During the same year, 50 existing TB patients are already on treatment. What is the incidence rate of TB in that district?

- a) 25 per 100,000
- b) 100 per 100,000
- c) 125 per 100,000
- d) 75 per 100,000

Q133. A 35-year-old migrant worker presents with cough for 3 weeks, evening rise of temperature, and weight loss. Sputum smear microscopy is positive for acid-fast bacilli. According to the National Tuberculosis Elimination Programme (NTEP), what is the next step

- a) Start ATT as 2HRZE + 4HRE
- b) Start 6 month BPaLM
- c) Start 6 month BPaL
- d) FIRST do CBNAAT and start treatment based on rifampicin sensitivity

Q134. A 2-year-old child from a rural area presents with cough and fast breathing for 2 days. On examination, respiratory rate is 48/min, chest indrawing is present, and there is no stridor. The child is alert and feeding properly. According to IMNCI guidelines, how should this child be classified?

- a) No pneumonia
- b) Pneumonia
- c) Severe pneumonia
- d) Very severe disease



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Q135. A 45-year-old woman working as a cook in a school hostel has a history of typhoid fever 2 years ago. Stool culture repeatedly shows *Salmonella typhi* but she is asymptomatic. Which of the following best explains her epidemiological significance?

- a) She represents a convalescent carrier
- b) She is an incubatory carrier
- c) She is a chronic carrier
- d) She is a healthy carrier

Q136. A 32-year-old male from a coastal village presents with profuse watery diarrhoea described as “rice-water stools” and severe dehydration. Over the next 48 hours, 27 similar cases are reported from the same locality. The medical officer suspects cholera outbreak. Which is the most appropriate immediate public health measure to prevent further transmission?

- a) Mass vaccination with oral cholera vaccine
- b) Isolation of all cases in tertiary hospital
- c) Super chlorination of drinking water source
- d) Prophylactic antibiotics to all contacts

Q137. In a community-based survey, 5% of antenatal women tested positive for HIV in a state. This state is classified as:

- a) Low prevalence state
- b) High prevalence state
- c) Moderate prevalence state
- d) Concentrated epidemic area

Q138. A 19-year-old male presents with urethral discharge and dysuria. Gram stain shows gram-negative intracellular diplococci. As per national guidelines, incorrect about this case



- a) Treated as syndromic approach under Suraksha clinic
- b) Grey kit- 1
- c) Brown kit-8
- d) Treated with azithromycin and Cefixime

Q139. A 28-year-old dairy worker presents with intermittent fever for 3 weeks associated with night sweats and joint pains. He consumes unpasteurized milk regularly. Most likely diagnosis?

- a) Anthrax
- b) Brucellosis
- c) Q fever
- d) Leptospirosis



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- Q140. A 40-year-old paddy field worker presents with fever, severe myalgia (especially calf muscles), conjunctival suffusion, and jaundice. He gives history of working in flooded fields. What is the most likely organism?
- Leptospira interrogans
 - Hantavirus
 - Brucella melitensis
 - Rickettsia tsutsugamushi
- Q141. A 32-year-old shepherd from a hilly area presents with sudden high-grade fever, chills, and painful swelling in the groin. Examination reveals a tender enlarged inguinal lymph node. He reports handling dead rodents in the past week. Most likely diagnosis?
- Tularemia
 - Bubonic plague
 - Leptospirosis
 - Scrub typhus
- Q142. A 7-year-old boy is brought to the PHC after being bitten on the face by a stray dog while playing. The wound is bleeding and multiple deep puncture marks are seen. The dog ran away and its vaccination status is unknown. The child has never been vaccinated against rabies. What is the most appropriate management?
- Wash wound and give anti-rabies vaccine only
 - Give rabies immunoglobulin only
 - Wash wound + anti-rabies vaccine + rabies immunoglobulin
 - Observe the dog for 10 days before starting treatment
- Q143. Under NVBDCP, a village reports 5 confirmed malaria cases in a week. Blood examination rate (ABER) is below 5%. What does low ABER indicate?
- High transmission
 - Good surveillance
 - Poor surveillance
 - Vector resistance
- Q144. A district reports rising dengue cases during monsoon. Entomological survey shows Aedes aegypti Index of 15% and Breteau Index of 25%. Which of the following is the most appropriate immediate public health action?
- Fogging only
 - Mass chemoprophylaxis
 - Source reduction and larval control
 - Isolation of all patients
- Q145. A 6-year-old child from rural Uttar Pradesh presents with high fever, seizures, altered sensorium. MRI shows thalamic involvement. Pigs are reared near his house. Incorrect about above disease:
- Vector mosquito breed in artificially collected water
 - Pig- amplifier host



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- c) Vaccine preventable disease and vaccine is part of UIP program
- d) Mosquito vector- culex

Q146. A 40-year-old man from coastal Odisha presents with progressive swelling of lower limbs and scrotum for 5 years. Peripheral smear collected at 2 AM shows microfilariae. All are correct about disease except

- a) Single thick blood smear for diagnosis
- b) Ivermectin, albendazole and dec used for treatment
- c) Annual mass drug administration of ivermectin, albendazole and dec in area with microfilaremia is < 1%
- d) Vector is culex

Q147. A 32-year-old man from Bihar presents with prolonged fever (>3 months), weight loss, hyperpigmentation of skin, and massive splenomegaly. rk39 test is positive. Correct about treatment:

- a) Lyposomal amphotericin B single dose I/V infusion as drug of choice
- b) Lyposomal amphotericin B single dose oral as drug of choice
- c) Miltefosine single dose I/V infusion as drug of choice
- d) Miltefosine single oral as drug of choice

Q148. During a cholera outbreak in a district, 200 cases were reported. Out of these, 20 patients died. What is the case fatality rate (CFR)?

- a) 5%
- b) 10%
- c) 15%
- d) 20%

Q149. A malaria outbreak occurred after heavy rainfall in a rural village. Stagnant water led to mosquito breeding. Many non-immune migrant workers developed fever. The outbreak can best be explained by disturbance of:

- a) Web of causation
- b) Epidemiological triad
- c) Bradford Hill criteria
- d) Koch's postulates

Q150. A state health officer reviews monthly immunization coverage reports and checks whether targets are being met during program implementation. This activity is an example of:

- a) Evaluation
- b) Supervision
- c) Monitoring
- d) Audit



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ANSWERS

1	c
2	c
3	a
4	c
5	a
6	a
7	b
8	a
9	b
10	b
11	c
12	a
13	b
14	b
15	b
16	b
17	b
18	a
19	c
20	b
21	b
22	b
23	b
24	c
25	c
26	b
27	b
28	b
29	b
30	a

31	c
32	c
33	b
34	c
35	c
36	b
37	b
38	b
39	d
40	b
41	b
42	a
43	b
44	a
45	b
46	a
47	b
48	c
49	b
50	a
51	b
52	c
53	c
54	c
55	c
56	c
57	b
58	c
59	b
60	b

61	b
62	b
63	b
64	b
65	d
66	b
67	b
68	c
69	b
70	b
71	b
72	c
73	b
74	c
75	a
76	c
77	c
78	a
79	c
80	c
81	b
82	b
83	b
84	a
85	a
86	b
87	b
88	b
89	b
90	b

91	c
92	b
93	a
94	b
95	a
96	b
97	c
98	b
99	c
100	b
101	b
102	c
103	b
104	b
105	b
106	b
107	b
108	c
109	b
110	b
111	c
112	c
113	a
114	b
115	c
116	b
117	c
118	b
119	b
120	b

121	c
122	d
123	d
124	c
125	c
126	b
127	b
128	c
129	b
130	b
131	b
132	b
133	d
134	b
135	c
136	c
137	b
138	c
139	b
140	a
141	b
142	c
143	c
144	c
145	a
146	c
147	a
148	b
149	b
150	c